AI-Powered Robo-Advisor Platform

Academic Project Report
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Contents

1 Introduction

Most commercial robo-advisors share three structural limitations: opaque models, undisclosed training data, and unconstrained AI components. This project addresses all three in an academic prototype combining a Gradient Boosting risk profiler trained on the Federal Reserve Survey of Consumer Finances (SCF 2022), a Hierarchical Risk Parity (HRP) portfolio optimizer with Ledoit–Wolf covariance shrinkage, and a constrained LLM narrator (large language model API) that is mathematically prevented from inventing figures via a Ground Truth JSON contract and a five-step validator. An EU Awareness layer (Rule 9) explicitly communicates the US-centric bias of the SCF training data to European investors at the point of recommendation.

The platform is a Streamlit web application backed by a FastAPI service, with a SQLite audit trail storing SHA-256 hashes of every market data snapshot and LLM prompt. An agentic GitHub Actions workflow demonstrates the mandatory AI agent criterion: a pull request generated entirely by the LLM API without human intervention.

Source code and Architecture Decision Records: `https://github.com/Programming-for-finance-II/robo-advisor`

The repository also contains two supplementary documents submitted alongside this report: `docs/development_plan.pdf` (project plan, timeline and role allocation) and `docs/project_diary.pdf` (full session-by-session development diary, covering every work session from 27 April to 12 June 2026 across all four roles).

2 Backend and Data Infrastructure (P1)

2.1 API Layer

The backend is a **FastAPI** service exposing five endpoints: `/profile` (risk profiler), `/optimize` (portfolio optimiser), `/advice` (LLM narrator), `/backtest` (historical scenario analysis), and `/compare` (HRP vs. MVO side-by-side). All protected endpoints require an `X-API-Key` header validated on every request. Rate limiting is enforced via `slowapi` per IP address (20 requests per minute on `/profile`, 10 on the costlier `/advice`); requests exceeding the limit receive a `429 Too Many Requests` response without consuming any computation. All responses are Pydantic-validated, ensuring schema consistency between the backend and the Streamlit frontend.

2.2 Data Pipeline: ValidatedDataLoader

Market data is downloaded from `yfinance` by the **ValidatedDataLoader** (`backend/data/loader.py`). The loader applies three sequential quality controls before returning data to the optimiser:

1. **UCITS probe.** For each primary UCITS ticker, a lightweight download is performed and the per-ticker NaN ratio is computed. If the ratio exceeds the 2 % threshold, the loader substitutes the corresponding US-listed fallback (e.g. `SPY` for `CSPX.L`) and records the substitution.
2. **Forward-fill.** Gaps of up to two consecutive trading days are filled; remaining NaN values cause a `DataQualityError`.
3. **Minimum observations gate.** At least 252 valid trading days (one calendar year) must remain after cleaning; shorter series are rejected.

The cleaned price matrix is serialised to CSV and hashed (SHA-256) to produce the `market_data_hash` stored in the audit trail.

2.3 Audit Trail

The SQLite database (`robo_advisor.db`) contains two tables. `market_data_snapshots` stores one row per unique `market_data_hash`, using `INSERT OR IGNORE` for deduplication: identical market data downloaded on different days produces a single stored snapshot. `recommendations` stores one row per optimisation request, linking to the corresponding snapshot via the hash and recording the full input parameters, profile output, portfolio weights, and risk metrics. This design enables bit-for-bit reproducibility of any historical recommendation from the stored hash alone. The application is containerised via `docker-compose.yml`; on Streamlit Community Cloud the database resets on redeploy, as documented in ADR-005.

3 ML Risk Profiler (P3)

3.1 Architecture

The profiler combines two components behind a shared interface. The **rule-based engine** implements the Grable–Lytton (1999) instrument (Grable and Lytton 1999): ten questionnaire items scored on $[0, 30]$ are mapped to three profiles (`CONSERVATIVE`, `MODERATE`, `AGGRESSIVE`), with a capital-preservation override that caps the profile regardless of total score; this deterministic, MiFID II-aligned engine is the live serving path. Alongside it, a **Gradient Boosting classifier** trained and validated on empirically observed household behaviour provides an empirically grounded model. Both return a uniform `ProfilerOutput` typed dictionary, so downstream components are agnostic to which one produced the classification.

3.2 Label Construction and GBM Classifier

Training labels are derived from the SCF 2022 (Board of Governors of the Federal Reserve System 2022) ($n = 4{,}595$ households, implicate 1). Four allocation variables (`EQUITY`, `BOND`, `CASHLI`, `STOCKS`) are normalised and clustered via K-Means ($K = 3$, silhouette $s = 0.84$). Clusters are labelled deterministically by mean equity share. A `HistGradientBoostingClassifier` is trained on nine questionnaire-derived features with SCF sample weights preserved. Performance:

Metric	HistGBM	LR Baseline
Training accuracy	97.7%	79.9%
CV accuracy (3-fold)	94.0% ± 0.15%	63.3% ± 2.9%

Table 1: Classifier performance on SCF 2022.

3.3 SHAP Explainability

SHAP values are computed via `TreeExplainer` and normalised by absolute magnitude. The top-N drivers are injected into the Ground Truth JSON; the LLM narrator is constrained to reference only features present in `top_drivers` (Rule 7), preventing fabricated correlations.

4 Portfolio Optimisation (P2)

4.1 ETF Universe

The platform allocates across eight ETFs providing multi-asset diversification under MiFID II constraints, with three UCITS-compliant primary tickers and US-listed fallbacks for the remainder.

Table 2: ETF universe — primary tickers, clusters, UCITS status.

Ticker	Asset class	Cluster	UCITS
CSPX.L	Equity USA	risk_assets	✓
EFA	Equity International	risk_assets	
GLD	Gold	real_assets	
VNQ	REIT USA	real_assets	
AGGH.MI	Bond Aggregate (EUR)	safe_haven	✓
TLT	US Treasury 20Y+	safe_haven	
TIP	TIPS Inflation-Linked	safe_haven	
XEON.MI	Cash / Overnight	cash	✓

4.2 Hierarchical Risk Parity

HRP (López de Prado 2016) was selected over MVO for three structural reasons (ADR-001): it requires no matrix inversion, no expected-return estimate, and produces no corner solutions. The algorithm proceeds via Ward-linkage tree clustering on a Ledoit–Wolf distance matrix, quasi-diagonalisation, and recursive bisection assigning weights inversely proportional to intra-cluster variance:

$$\alpha_L = 1 - \frac{V_L}{V_L + V_R}, \qquad V_k = \mathbf{w}_k^\top \hat{\Sigma}_k \, \mathbf{w}_k$$

Raw weights are tilted by risk profile: Conservative blends toward Minimum Variance; Aggressive toward Equal Risk Contribution (Maillard, Roncalli, and Teïletche 2010); Balanced uses raw HRP weights.

4.3 Ledoit–Wolf Shrinkage and Constraints

The Ledoit–Wolf shrinkage estimator (Ledoit and Wolf 2004) $\hat{\Sigma} = (1-\alpha)S + \alpha\mu_S I$ is applied before clustering to stabilise the distance matrix. Box constraints clip weights to $[5\%, 40\%]$ per asset and $[10\%, 60\%]$ per cluster; weights are renormalised after clipping.

4.4 Market Regime Detection

A dual-trigger detector flags `HIGH_STRESS` when $\bar{\rho}_{\mathrm{LW}} > 0.75$ or VIX > 30 (ADR-006). When triggered, the frontend surfaces a warning banner cautioning that diversification benefits are reduced in the prevailing regime. A cluster-level Equal Risk Contribution allocation (`get_erc_cluster_weights`) is implemented as the defensive response for this regime; wiring it into the automatic serving path is left as future work.

5 LLM Narrator and Validator (P4)

5.1 Ground Truth JSON Contract

All numerical computations are performed by the backend. The LLM receives a `GroundTruthPayload` containing pre-computed figures and is permitted only to narrate them. The payload includes portfolio weights, risk metrics (volatility, max drawdown, VaR, CVaR), profiler output with SHAP drivers, an `allowed_numbers` whitelist (auto-populated by recursively extracting every float from the payload), and regulatory context. `expected_annual_return` and `sharpe_ratio` are intentionally `null` for HRP, as the algorithm produces no reliable forward-return estimate.

5.2 System Prompt: Nine Absolute Rules

The narrator operates under nine rules enforced at prompt level: (1) no invented numbers outside `allowed_numbers`; (2) no prescriptive advice (forbidden phrases injected from the payload); (3) no compliance claims or absolutes; (4) historical framing only for return figures; (5) fixed out-of-context response; (6) cluster descriptions in economic, not correlation, terms; (7) SHAP drivers only; (8) mandatory disclaimer; (9) **EU Awareness** — when `profiler_us_centric_caveat` is true, the narrator must acknowledge SCF US-centrism and European investor differences.

5.3 Five-Step Validator

Every response passes through `validator.py` before reaching the user. Steps 1, 2, 4, 5 are blocking; Step 3 is corrective:

1. **Forbidden phrase check** — case-insensitive substring scan.
2. **Hallucinated number check** — all floats extracted via regex, percentages normalised, checked against `allowed_numbers` within $\pm 2\%$ tolerance.
3. **Disclaimer presence** — auto-appended if absent.
4. **Injection detection** — semantic scan of output for known injection patterns.
5. **EU Awareness (Rule 9)** — verifies SCF/Federal Reserve reference and European investor reference are both present when required.

5.4 Prompt Injection Defence

Layer 1 sanitises user input before the API call: inputs exceeding 500 characters are rejected; known injection keywords are blocked case-insensitively. Accepted input is wrapped in `<user_input>` tags. Layer 2 is Validator Step 4, catching cases where the model echoes injected instructions in its output.

6 Backtest Results

Seven historical episodes are evaluated for the MODERATE profile: five crises (Global Financial Crisis 2008, Eurozone debt crisis 2011, COVID-19 crash 2020, Ukraine invasion shock 2022, rate-hike cycle 2022), one moderate-stress year (rate-fear selloff 2018), and one positive regime (post-COVID bull 2021). Each backtest rebalances monthly on a 252-day lookback with 10 bps of transaction costs per rebalance. The three UCITS primaries lack price history for the earlier windows, so the engine substitutes US-listed proxies (`SPY`, `AGG`, `BIL`); cluster box constraints are applied to these proxies under their primaries' clusters, so the profile cap on cash binds throughout the historical window (an earlier version mismatched the constraint keys to the proxy tickers, silently leaving $\sim$40 % in cash and understating drawdowns).

Table 3: Backtest summary — HRP vs. MV vs. $1/N$ (MODERATE profile).

Scenario	Model	Ann. Return	Volatility	Sharpe	Max DD
GFC (2008)	HRP	−5.4%	16.0%	−0.33	−22.3%
	MV	−9.3%	11.8%	−0.78	−21.5%
	$1/N$	−4.9%	16.6%	−0.29	−23.2%
Eurozone (2011)	HRP	+5.7%	6.0%	+0.95	−4.0%
	MV	+9.3%	8.1%	+1.15	−6.9%
	$1/N$	+7.8%	8.8%	+0.89	−5.5%
Selloff (2018)	HRP	−2.3%	4.2%	−0.55	−4.8%
	MV	−3.7%	8.6%	−0.43	−7.1%
	$1/N$	−3.4%	5.7%	−0.60	−6.7%
COVID-19 (2020)	HRP	+7.5%	9.7%	+0.77	−13.9%
	MV	+6.4%	11.7%	+0.55	−15.0%
	$1/N$	+10.9%	11.2%	+0.97	−15.0%
Bull (2021)	HRP	+5.3%	3.5%	+1.53	−2.2%
	MV	+7.7%	8.3%	+0.94	−8.1%
	$1/N$	+8.7%	5.3%	+1.65	−3.3%
Ukraine (2022)	HRP	−19.2%	8.5%	−2.26	−10.3%
	MV	−16.7%	13.5%	−1.24	−10.5%
	$1/N$	−22.5%	11.5%	−1.97	−12.5%
Rate Hike (2022)	HRP	−11.2%	9.2%	−1.21	−16.1%
	MV	−10.6%	10.1%	−1.05	−13.9%
	$1/N$	−13.7%	11.9%	−1.15	−19.6%

No single strategy dominates across all seven episodes, and the broadened coverage makes the trade-offs explicit. HRP records the lowest volatility in five of the six episodes — the 2008 crisis is the exception, where the Minimum-Variance-tilted MV is lower — and the smallest maximum drawdown in the asymmetric crises where diversification matters most, most clearly the 2011 Eurozone debt crisis, where it stays positive (+5.7%) with only a −4.0% drawdown while European equities fell. In the sharpest equity selloffs (2008 and 2022), however, MV achieves a smaller drawdown than HRP, and in the recovery and bull phases (2020, 2021) the naive $1/N$ portfolio earns both higher returns and a higher Sharpe ratio. HRP's contribution is therefore risk control and robustness across regimes rather than uniform outperformance — consistent with its construction, which targets diversification rather than return maximisation.

7 Limitations

yfinance fragility. The platform relies on an unofficial API with no SLA. UCITS-listed tickers occasionally return excessive NaN values; the `ValidatedDataLoader` mitigates this with automatic fallback, but live operation remains dependent on third-party data availability.

SCF US-centrism. The risk profiler is trained on US household data. European investors may exhibit materially different risk preferences, savings rates, and financial literacy patterns. The EU Awareness layer communicates this explicitly at the point of recommendation, but does not correct the underlying distributional mismatch.

HRP opacity. The recursive bisection algorithm produces weights that cannot be summarised by a simple rule. The LLM narrator and the MVO comparison tab partially mitigate this, but residual opacity is a fundamental limitation acknowledged in ADR-001.

Validator false positives. The substring check for *safe* also matches the cluster label *safe_haven*, triggering unnecessary fallback responses. Whole-word matching is deferred to future work. The EU Awareness check (Step 5) is keyword-based rather than semantic and may fail on paraphrased compliant responses.

SQLite persistence. The audit trail resets on Streamlit Cloud redeploy. A managed PostgreSQL instance would be required in production (ADR-005).

8 Lessons Learned

Process over product: the agentic workflow. The mandatory AI-agent criterion was satisfied by a GitHub Actions workflow that opens pull requests autonomously through the LLM API (PR #43). The main obstacle was infrastructural: in a private organisation the default `GITHUB_TOKEN` cannot open pull requests, which we resolved with a scoped Personal Access Token. Pairing branch protection with mandatory CI turned the agentic process into a reviewable, auditable record rather than an opaque automation — precisely the "process over product" the brief asks for.

Architecture as a safety mechanism. The central design choice — a strict separation between a deterministic "calculator" (the backend) and an LLM "narrator" constrained to a Ground Truth JSON contract — made hallucination prevention *structural* rather than instructional. The narrator is stateless: the payload is re-injected on every call, so the model stays anchored to current data and cannot drift across turns. This proved far more robust than relying on prompt wording alone.

Tests and CI catch real bugs. A covariance-frequency bug (`CovarianceShrinkage(frequency=1)` without an explicit annualisation factor) would have inflated reported volatility by $\sqrt{252} \approx 15.9\times$; it surfaced only because the risk-metrics layer annualises explicitly and a regression test pinned the expected value. Smaller edge cases (e.g. integer-valued floats slipping through the validator's number check) emerged the same way. Coverage was governed through a shared GitHub issue rather than by unilaterally excluding untested modules, lifting the project from 77 % to 81 %.

Generative AI tools used. All four team members used generative AI throughout the project, as required by the course brief. **Claude (Anthropic)** was the primary coding assistant and technical advisor across all weeks; it also powers the Chat Advisor at runtime via the Claude API. **ChatGPT (OpenAI)** was used for initial brainstorming and for explaining financial concepts (HRP, MiFID II, SCF survey methodology). **GitHub Copilot / Gemini** provided code comparison and alternative implementation suggestions during development. All AI-assisted contributions are visible in the commit history; the agentic criterion is satisfied by `agent_pr.yml` (PR #43).

Collaboration, honesty and reproducibility. Parallel work across four roles produced classic distributed-Git frictions: long-lived branches drifting tens of commits behind `main`, cross-team review of optimiser fixes, and the consolidation of every schema into a single `backend/schemas` source of truth. A pre-publication secrets audit (scanning the full history for API-key patterns before the repository was made public) and an honest accounting of what is and is not wired into the live serving path — the trained GBM and the ERC stress fallback are implemented but not on the live path — were treated as part of the deliverable, not an afterthought.

9 Conclusions

This paper presented an educational robo-advisor prototype demonstrating a complete investment workflow — from risk profiling to portfolio optimisation to natural-language explanation — while explicitly addressing transparency and geographic-awareness gaps common in commercial platforms. The Ground Truth JSON contract and the five-step validator represent the core architectural innovation: they make hallucination prevention structural rather than instructional. The EU Awareness layer (Rule 9, `profiler_us_centric_caveat`, Validator Step 5) is a deliberate design choice to surface the geographic limitations of a US-trained model to European investors at the point of recommendation.

Future Work

Replace the SCF 2022 training dataset with the ECB Household Finance and Consumption Survey (HFCS); replace `yfinance` with a reliable data provider; implement whole-word matching in the validator; add narrator retry logic before returning the safe fallback; wire the cluster-level ERC fallback into the live serving path under `HIGH_STRESS`; extend the regime detector with a rolling confirmation window.

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